UNIVERSIDAD DE LA SABANA

Big Data Tools – Coterminal Ingeniería Informática.

Tutorial 1: Configure Docker Desktop

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Windows 11

- Download Docker Desktop from the official Docker website for Windows
    - https://www.docker.com/products/docker-desktop/
- Prerequisites:
- As administrator, enable Windows Subsystem for Linux. Run Terminal or PowerShell in Admin mode and run:
    - dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
- As administrator, enable Virtualization support:
    - dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
- Enable Windows Subsystem for Linux 2 (WSL 2) by running wsl --install in PowerShell as Administrator, if not already enabled.
    - *Windows Professional and BIOS settings could be required*
- Run the Docker Desktop Installer (.exe) after download, follow the on-screen instructions.
- During installation, select the backend as WSL 2 (recommended) or Hyper-V based on system support.
- After installation, restart your computer if prompted.
- Optionally, add your user to the docker-users group if using different admin and user accounts.
- Verify installation by running in Terminal or PowerShell terminal:

```
docker --version
```

```
docker run hello-world
```

Docker Desktop integrates tightly with WSL 2 on Windows 11 for best performance and development experience.

macOS Sonoma and Previous Recent Distributions

- Download Docker Desktop for Mac from Docker's official website.
- Supports both Apple Silicon (M1, M2, M3) and Intel-based Macs.

- Installation involves opening the downloaded DMG file, dragging the Docker icon to the Applications folder, and launching Docker from there.
- Follow the on-screen prompts to accept subscription terms (Docker Desktop is free for educational use).
- After installation, verify Docker is running by opening a terminal and running:

```
docker --version
```

```
docker run hello-world
```

This method works well with macOS Sonoma and includes support for both Intel and Apple Silicon Macs.

Linux Developer Environment

- For Linux, you typically install Docker Engine (Docker CE) through your distribution's package manager. Installation instructions vary by distro (Ubuntu, CentOS, Fedora, etc.) but are well documented.
- Alternatively, Docker Desktop for Linux is available and provides a GUI similar to the Windows and macOS versions.
- Docker Desktop for Linux runs inside a lightweight VM and manages Docker contexts separately from Docker Engine.
- To avoid port conflicts, it is recommended to stop the traditional Docker Engine service while using Docker Desktop on Linux.
- Commands to stop and disable Docker Engine service may be:

```
bash
```

```
sudo systemctl stop docker docker.socket containerd
```

```
sudo systemctl disable docker docker.socket containerd
```

- Verify installation by running:

```
bash
```

```
docker --version
```

```
docker run hello-world
```